

FR + HHT Update Report (Alpha & Skymap/Chirp) — 1V0

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Executive Summary

- IMF2 HHT 'locking' favors $\ln(1+z)$ with $\alpha = 1.0$ over $\alpha = 0.5$ based on paired comparisons (mean $|r|$: 0.909 vs 0.879; identical median p owing to the test floor).
- Skymap/locking heatmaps from the 2-bin BAO pack (DH_over_rd, DM_over_rd) are consistent with $\alpha = 1.0$.
- FR + OU(Λ) with amp = $1e-120$ is robust to corrz (0.05–0.60) at fixed seed; $\chi^2/\text{dof} \approx 1.166$ (SN-only).
- Piecewise Λ : discrete-events $\Delta z = 1e-4$ matches continuous ($\chi^2 \approx 1975.21$), validating the integrator.
- PBH with $\Lambda \leftrightarrow A_{\text{acc}}$ coupling $\gamma \in \{0.00, 0.02, 0.05, 0.10\}$ yields identical SN-only best fits ($\chi^2 \approx 1974.51$).
- SN+BAO FR with $c_{\text{of_z}}$ slightly improves χ^2 vs c_0 (2014.55 vs 2017.30) under spline- Λ fluctuations.
- Soft-link $EM \leftrightarrow c$ (flat FR, $\log_{10} p_z$) with $\kappa \approx -1$, $\sigma = 0.10$ increases χ^2 (~ 2005.95) — currently tension-adding.

HHT Locking & Alpha Comparison

- Tools: hht_locking_heatmap.py; hht_alpha_compare_quick.py; hht_alpha_compare_report.py (1V0).
- CEEMDAN, IMF2, seeds {3,7,11}, ns=200 $\rightarrow \alpha=0.5$ win-rate 0% vs $\alpha=1.0$ on $|r|$ and p ; mean $|r|$ 0.879 vs 0.909.
- Interpretation: IMF2 locking favors $\ln(1+z)$ ($\alpha=1$).

Skymap / Chirp Notes

- HHT locking heatmaps ('skymaps') show stronger IMF2 ridge coherence along $\alpha=1.0$ axis.
- Next: extend to IMF3 for chirp energy tracking and BAO (DH/DM) stability vs seeds/method.

FR — OU(Λ) Robustness & Amplitude

- FR + OU(Λ), amp= $1e-120$; corrz $\in \{0.05, 0.15, 0.30, 0.60\}$; emcee: walkers=20, steps=700, burn=200.
- Best fit repeats across corrz: $H_0 \approx 82.65$, $\Omega_m \approx 0.575$, $\Omega_\Lambda \approx 0.964$, $A_{\text{acc}} \approx -0.250$, $n_{\text{acc}} \approx -5.91$, $\gamma_{\text{c}} \approx +0.053$, $\varepsilon_M \approx +0.255$.
- Amplitude sweep (no-MCMC): unchanged up to $1e-70$; at $1e-2$ χ^2 degrades to ≈ 2089.88 .

FR — Piecewise Λ : Continuous vs Discrete

- Piecewise Λ (knots=8): continuous $\chi^2_{\text{SN}} \approx 1975.24$; discrete $\Delta z = 0.015$ $\chi^2_{\text{SN}} \approx 1980.43$; $\Delta z = 1e-4$ (at continuous best-fit) $\chi^2_{\text{SN}} \approx 1975.21$.
- Conclusion: discrete formulation converges to the continuous limit as $\Delta z \rightarrow 0$.

PBH — $\Lambda \leftrightarrow A_{\text{acc}}$ Coupling Sweep

- PBH + OU(Λ), amp= $1e-120$, corrz=0.25, $\Lambda \leftrightarrow A_{\text{acc}}$ coupling $\gamma \in \{0.00, 0.02, 0.05, 0.10\}$.
- All runs converge to the same SN-only best fit: $\chi^2 \approx 1974.51$; suggests γ is unconstrained without BAO/Planck.

SN + BAO (FR, spline- Λ)

- FR + BAO (spline- Λ): $c_0 \rightarrow \chi^2_{\text{total}} \approx 2017.30$; $c_{\text{of_z}} \rightarrow \chi^2_{\text{total}} \approx 2014.55$.
- Both above SN-only best levels; $c_{\text{of_z}}$ is modestly better.

EM $\leftrightarrow c$ Soft Link Tests

- Soft link $EM \leftrightarrow c$ (flat FR, $dM = \log_{10} p_z$, $c = \log_{10} p_z$): $\kappa \approx -1$, $\sigma = 0.10 \rightarrow \chi^2_{\text{total}} \approx 2005.95$.
- Grid over (κ, σ) pending to quantify $\Delta\text{AIC}/\Delta\text{BIC}$ and find a sweet-spot prior.

Paths & Artefacts

- /Users/boyde/.spyder-py3/plamb_runs/tools/hht_imf2_compare/locking_alpha_compare.csv
-
- /Users/boyde/.spyder-py3/plamb_runs/tools/hht_imf2_compare/HHT_Locking_Alpha_Compare_1V0.docx
- /Users/boyde/.spyder-py3/plamb_runs/SNBAO_baocmode_c0/FR_emcee_bestfit.txt

- /Users/boyde/.spyder-py3/plamb_runs/SNBAO_baocmode_c_of_z/FR_emcee_bestfit.txt
- /Users/boyde/.spyder-py3/plamb_runs/SN_OU_amp1e-120_corr0.30/FR_emcee_bestfit.txt
- /Users/boyde/.spyder-py3/plamb_runs/SN_piecewise_cont/FR_emcee_bestfit.txt
- /Users/boyde/.spyder-py3/plamb_runs/SN_piecewise_disc/FR_emcee_bestfit.txt
- /Users/boyde/.spyder-py3/plamb_runs/SN_piecewise_disc_at_cont/FR_none_bestfit.txt
- /Users/boyde/.spyder-py3/plamb_runs/SN_piecewise_disc_at_cont_DZ1e-4/FR_none_bestfit.txt
- /Users/boyde/.spyder-py3/plamb_runs/PBH_couple_gamma_0.00/PBH_emcee_bestfit.txt

Next Steps

- Add Planck distance priors (planck_c_mode $\in$ {none,R_only,R_and_rs}).
- Run IMF3 locking + chirp ridge detection; document surrogate significance.
- SN+BAO: vary rs_power and c-model (log1pz, piecewise, saturating).
- Formalize (κ, σ) soft-link grid; compare via $\Delta AIC/\Delta BIC$.
- Automate CSV→DOCX tables; embed corner/traces from each run.